

Data-Driven Marketing Analytics Dashboard
3-Month Internship Project | Domain: Data Analytics

Project Summary

This project builds an end-to-end marketing analytics solution that tracks campaign performance, measures ROI and generates actionable insights across 5 digital marketing channels (Google Ads, Facebook Ads, Instagram Ads, Email, SEO Organic).

Key Results:

- 50 campaigns analysed across FY 2024
- 12.5M impressions | 853K clicks | 37.8K conversions
- Total Revenue: \$1,186,390 | Total Cost: \$308,779 | ROAS: 3.84x
- Email: Highest ROAS (9.25x) | Google Ads: Highest revenue (64% share)

Project Structure

...

```
marketing_analytics_project/
├── data/
│   ├── campaigns.csv          # 50 campaigns, 18 attributes
│   ├── daily_performance.csv  # 2,486 daily observations
│   ├── kpi_summary.csv        # Channel KPI aggregations
│   └── sql_results/           # Query result CSVs (6 files)
│       ├── 01_overall_kpi.csv
│       ├── 02_channel_kpis.csv
│       ├── 03_top10_campaigns.csv
│       ├── 04_campaign_type_perf.csv
│       ├── 05_monthly_trend.csv
│       └── 06_budget_recommendation.csv
├── scripts/
│   ├── generate_data.py       # Step 1: Generate simulated dataset
│   ├── eda_visualization.py   # Step 2: EDA & 10 chart figures
│   ├── run_sql_queries.py     # Step 3: Run SQL queries via SQLite
│   ├── build_dashboard.py     # Step 4: Build Excel dashboard
│   └── build_report.js        # Step 5: Build Word report (Node.js)
├── sql/
│   └── marketing_queries.sql   # 15 SQL queries with documentation
├── dashboard/
│   └── Marketing_Analytics_Dashboard.xlsx # 5-sheet Excel dashboard
├── reports/
│   ├── Marketing_Analytics_Report.docx   # Full project report
│   └── figures/                           # 10 chart PNG files
│       ├── 01_channel_roas.png
│       └── 02_revenue_vs_cost.png
```

```
├── 03_ctr_boxplot.png
├── 04_monthly_spend_trend.png
├── 05_conversion_funnel.png
├── 06_cpa_by_channel.png
├── 07_cost_vs_revenue_scatter.png
├── 08_heatmap_roas.png
├── 09_pareto_campaigns.png
├── 10_weekly_revenue_trend.png
└── README.md
```

...

Setup & Run Instructions

Prerequisites

```

**Python (3.8+)**
```bash
pip install pandas numpy matplotlib seaborn openpyxl
```

```

```

**Node.js (for report generation)**
```bash
npm install -g docx
```

```

...

Run the Full Pipeline

Execute scripts in order from the project root directory:

```
**Step 1 — Generate Data**
```bash
python scripts/generate_data.py
```
```

Output: data/campaigns.csv, data/daily_performance.csv, data/kpi_summary.csv

```

**Step 2 — Generate Visualisations**
```bash
python scripts/eda_visualization.py
```

```

Output: reports/figures/ (10 PNG charts)

```
**Step 3 — Run SQL Queries**
``bash
python scripts/run_sql_queries.py
``
```

Output: data/sql_results/ (6 CSV result files)

```
**Step 4 — Build Excel Dashboard**
```bash
```

```
python scripts/build_dashboard.py
```

```
'''
```

\*Output: dashboard/Marketing\_Analytics\_Dashboard.xlsx\*

**\*\*Step 5 — Build Project Report\*\***

```
```bash
```

```
node scripts/build_report.js
```

```
'''
```

Output: reports/Marketing_Analytics_Report.docx

Key KPIs Tracked

KPI	Formula	Value
CTR	Clicks / Impressions	6.83%
Conversion Rate	Conversions / Clicks	4.44%
CPC	Cost / Clicks	\$0.36 avg
CPA	Cost / Conversions	\$8.15 avg
ROAS	Revenue / Cost	3.84x
Profit	Revenue - Cost	\$877,611

Channel Rankings (by ROAS)

1. Email — 9.25x ROAS
2. SEO Organic — 6.08x ROAS
3. Google Ads — 4.14x ROAS
4. Facebook Ads — 2.95x ROAS
5. Instagram Ads — 2.60x ROAS

Tools & Technologies

- **Python: Pandas, NumPy, Matplotlib, Seaborn, openpyxl
- **SQL: SQLite (via Python sqlite3)
- **JavaScript / Node.js: docx library for Word report generation
- **Excel: openpyxl for dashboard automation
- **Power BI / Tableau: Dashboard design reference

Deliverables Checklist

- [x] Simulated marketing dataset (campaigns + daily)
- [x] 15 SQL analytical queries
- [x] 10 EDA visualisation figures
- [x] 5-sheet Excel dashboard with charts
- [x] Full project report (Word document)
- [x] README with setup instructions

